

11. Rating Political Candidates

A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating	Distance	d_1	d_2
1	12	75	3	4.5
2	7	165	1	-3
3	5	300	12	-1
4	19	15	1	11
5	17	180	8	2
6	12	240	11	-7.5
7	9	120	4	2
8	18	60	2	9
9	3	230	10	-9
10	8	200	9	-4
11	15	130	5.5	7.5
12	4	130	5.5	-7.5

- Calculate Spearman's rank correlation coefficient r_s .
- Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

$$r_s = 1 - \frac{6}{12(12-1)} \cdot 454$$

$$= -0.589$$

$$n=12, \alpha=0.05,$$

$$\text{臨界值} = 0.497$$

$$|-0.589| > |0.497|$$

→ reject H_0

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:⁸

Runner	Years of Competitive Running	10-Kilometer Finish Time (minutes)	d_1	d_2
1	9	33.15	1	6
2	13	33.33	2	7
3	5	33.50	3	0.5
4	7	33.55	4	2
5	12	33.73	5	3
6	6	33.86	6	-1
7	4	33.90	1	-5
8	5	34.15	8	-4.5
9	3	34.90	9	-8

- Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.

- Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = 0.05$.

$$r_s = 1 - \frac{6}{9(9-1)} \cdot 208.5$$

$$= -0.738$$

$$n=9, \alpha=0.05,$$

$$\text{臨界值} = 0.683$$

$$|-0.738| > 0.683$$

→ reject H_0